

Table2Table: Merging “Similar” Workspaces and Supporting Adaptive Telepresence Demonstration Guidance

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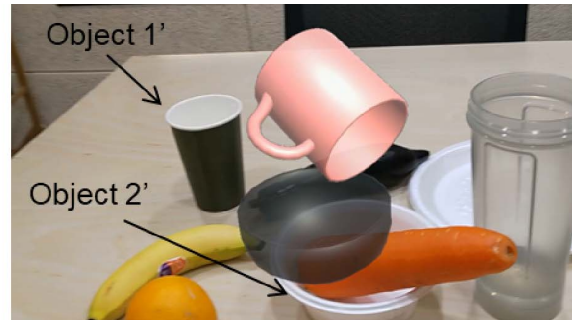
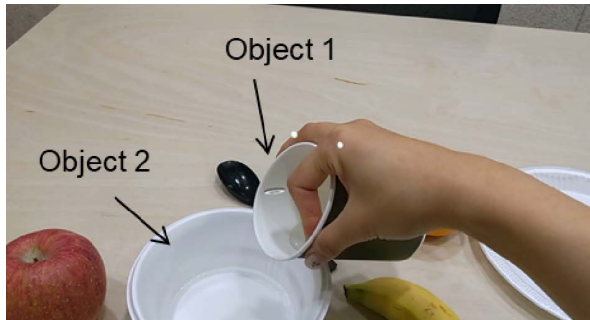


Figure 1: Table2Table: Instructor (in space A) pouring a cup (Object 1) of water in to a bowl (Object 2) (left), which is recognized and re-enacted in the student's (in space B) perspective with matched objects (Objects 1' and 2') in 3D animation of a cup being poured into another bowl (right).

ABSTRACT

In a typical tele-presence system, the remote participant is “transported” into the local space to be shared. However, such a set-up limits the remote participant in terms of physical interaction and demonstration with the objects in the local space. One alternative is to merge the two spaces, such that the student in one space can closely watch and mimic the actions of the instructor in its space - as often occurs in science laboratories or cooking classes. As the local set-ups in the two spaces may not be exactly the same, the student has to “adapt” to the situation and identify the right objects in mimicking the instructor. We propose to semi-automatically identify such object mappings, and based on the mapping, recognize the instructor’s action, and replicate it in the student’s space. Our preliminary user testing showed the possibility of the framework in improving the communication and process of learning.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

With the recent pandemic, people are becoming more accustomed to working from home by video conferencing, which is somewhat of a satisfactory means for simple meetings and communication, especially by only requiring a webcam. However, for social situations which require higher presence of the participants, and intimate and close collaboration, more advanced tele-presence systems will be needed [11] [3]. Typically, a tele-presence system captures the remote participant and transports it into another space, and allows ones to share and experience the space, and thereby has to employ an extensive set of sensors and displays (on both ends), and require an intricate set-up [14] [16]. Even so, such a system can only support simple “in person” non-physical communication. Tele-interacting

physically with the remote environment objects will be practically impossible.

Then, we pose the question of how to support effective tele-learning in a situation where hands-on physical demonstration with environment objects would be important, as often occurs in laboratory and cooking classes (see Figure 2).



Figure 2: A cooking class where the instructor demonstrates on one’s table and the students follow on their tables. The tools and materials (and their layout) on each table and used by the respective parties may not be exactly the same.

In this work, we present Table2Table, a tele-presence system, that supports such teaching based on hands-on demonstration - namely, the instructor manipulating and enacting on small objects on the table within the personal space, and the students trying to follow the same with one’s own objects (which may not be exactly the same as what the instructor uses).

We assume that in this hands-on demonstration, its visual enactment with the associated objects will be important (e.g. how exactly to stir the tomato sauce appropriately) which would be difficult to convey just through by mere aural instruction.

Since direct physical interaction with the objects in the local space is not possible (unless a tele-robotic haptic devices are used), the second-best solution might be to have the instructor demonstrate the routine on one’s own space (with one’s objects), capture the actions (e.g. instructor hand stirring the tomato sauce in the right way) as is (e.g. 3D video), and exporting and mapping it onto the student’s

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space, replayed and re-enacted in the first person viewpoint [10]. Achieving this in acceptable quality will require high-end sensors and object-level 3D reconstruction technology, and even so, the objects between the instructor's and student's space may not match (both spatially and semantically) [16] [14]. Students have to figure out how to follow the instructor's action and locate and match for the objects being manipulated. Other similar ways might be to play an instructional video on the student's side or capture video from the perspective of the instructor. The former is a relatively inexpensive and easy means but less effective, while the latter would require the instructor to wear a camera.

Therefore, we propose to, instead of as-is capture, recognize the instructor's action, identify the object mappings semi-automatically (using object recognition and similarity measure), and re-enact the demonstration in 3D graphics, applied to the mapped objects. Figure 1 illustrates the idea of our proposed system called "Table2Table (hence abbreviated as T2T)" - e.g. the instructor pours a liquid from a cup-1 of water into a bowl-1 (left), which is recognized and re-enacted in the student's perspective with the matched objects (e.g. bowl-1 to bowl-2, cup-1 to cup-2). Then, 3D animation of a "generic" cup being poured into another "generic" bowl is shown to the user with proper spatial registration. As the animation does not portray the instructor himself, the effect of the instructor presence is reduced (compared to the usual tele-presence system with captured remote participant). However, a separate live view of the instructor with voice instructions (easily captured) can be provided to supplement the animation, if needed. We conduct a user experiment to assess the effectiveness of our approach as compared to the usual live video conferencing approach.

2 RELATED WORK

One of the most attractive applications of the mixed reality (MR) is in the education/collaboration domain with its effectiveness demonstrated many times [1] [2] [6] [15] [9] [4]. Virtual reality (VR) offers learning situated and immersed in the target domain, freed from the physical constraints and limitations, e.g. learning about volcanoes around the virtual volcanic landscape without any physical danger [1]. Augmented reality (AR) based education can also provide effective learning by in-situ association of the real world objects with the relevant objects, useful information and even concepts [15] [2] [6].

Furthermore, tele-presence combined with MR based learning/collaboration can overcome the geographic differences, e.g. in learning about the French Revolution without having to actually visit the prison of Bastille in Paris. However, not many tele-presence MR based learning have considered hands-on "guidance" in which the instructor can "closely" help the students to enact certain physical routine, rather than just the distant visual demonstration.

A more advanced form of a tele-presence systems can capture and import the entire form/motion of the remote (instructor/collaborator) figure live into the local (student's) space [16] [14]. To this end, this type of system is focused on elevating the presence of the remote participant in the local space, but mostly limited to communicating with the remote participant in a non-physical way (e.g. voice/gesture conversation/instruction). Direct physical interaction with the objects in the local space would not be possible unless additional set-up with tele-robotic haptic device is in place.

The alternative non-physical method would be a demonstration which can be presented in various forms (e.g. 2D/3D video/animation, pre-recorded or live) and view points (e.g. 1st person, 3rd person, spatially registered). Depending on the form of the demonstration, the student may experience some difficulties as one has to match and find the proper objects to be manipulated in trying to follow what the instructor is doing.

Perhaps the best form of the non-physical demonstration would be to capture the instructor performing the routine on one's table

(naturally using his own objects) in 3D (as was done in [14] [5]), export and map it onto the student's space, and replay and re-enact it in the first person viewpoint. Implementation of such capabilities in acceptable quality (e.g. fine level spatial registration and augmentation, seamless 3D capture) will require high-end sensors and object-level 3D reconstruction technology, and even so, the objects between the instructor's and student's space may not match anyway (unless an exact same set-up is assumed). Thus, the students have to figure out how to follow the instructor's action and match for the objects being manipulated. In T2T, instead, we propose to establish the mapping between the instructor's objects to that of the student, recognize the actions (rather than capture), and re-enact it in the form of 3D animation as registered and augmented onto the mapped objects. This way, the cognitive load on both the instructor and the student are lessened.

While a typical tele-presence system with the participant (in space A) virtually transported to another (space B) cannot support physical interaction with the objects (of space B), one other way to achieve the intended effect is to have the two spaces replicated in the first place. This way, if the remote participant in space A acts on an object (in space A), the consequential outcome is replicated in the corresponding (replicated virtually or actually existent) object in space B. As such an assumption of two remote and local spaces to be exactly the same is difficult to uphold, few researches have explored how to achieve the same effect even when the two spaces are somewhat dissimilar. For example, Jo and Kim considered how to retarget the animated transported avatar to the dissimilar but corresponding objects in the other space (e.g. participant sitting on chair A in space A, transported to space B and having to sit on a somewhat dissimilar chair B) [7]. Yoon et al. considered a similar problem but to wider context, i.e. retargeting of the navigated path in one space to the other considering the spatial layout of the objects/obstacles in space and their semantics [18]. In our work, we tackle, as the object layouts and forms in the instructor's and student's table are not exactly the same, a similar problem - mapping the objects in space A to those in space B, so that the instructions unfolding in space A with objects in it can be demonstrated with the corresponding ones in space B.

3 TABLE2TABLE: SYSTEM CONFIGURATION

Figure 3 shows the overall system configuration of the Table2Table tele-presence system. The instructor is in space A, which encompass a table and cookery objects (e.g. bowl, cup, ...), and likewise for the student in space B. As indicated already, the object layouts and forms are not exactly the same, however, it is assumed that the student has all the necessary "matching" (that can fulfill the assigned role) objects to carry out whatever the instructor will do.

Space A is installed with a web camera and a monitor (which shows the scene from Scene B) facing the instructor. Note that one objective of T2T is minimizing complicated set-up as much as possible. In space B, the user wears an AR glass with a camera mounted on it (e.g. Hololens [12]) to view the scene with the augmented instruction animation. At first, the camera imagery from space A and space B are analyzed for existence, type and locations of the notable objects. The list of identified objects are initially matched based on their types, and presented to the student, and the student may revise the mapping if necessary. The system set-up in each space is depicted in Figure 4.

Once the mapping is established as such semi-automatically, the instructor will commence certain object manipulation routine, e.g. "pick a knife and cut a banana". Again, the incoming video of space A is analyzed with respect to the motion profiles of the detected hands and other objects to infer and recognize the action. The recognized action is to be replicated with the matched objects in space B as 3D animation in space B. Animations of the possible actions are made off-line, applied with the matched objects, registered to the image

space location of the matched objects, and presented in 1st person viewpoint through the AR glass. As an option (although not used in the current pilot experiment), the student may be also augmented with the video window which shows the view from the camera of space A (mirrored scene of the instructor's table). We provide further details of the individual modules in the overall system.

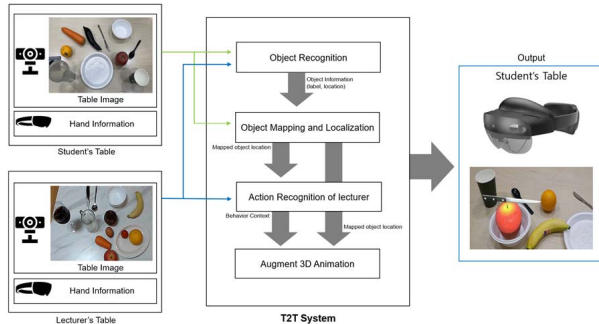


Figure 3: The overall system configuration of Table2Table. First the objects between the two instructor's and student's table spaces are semi-automatically established, then the instructor's actions recognized. The recognized action as applied to the matched objects is re-enacted in 3D animation in the student's perspective.

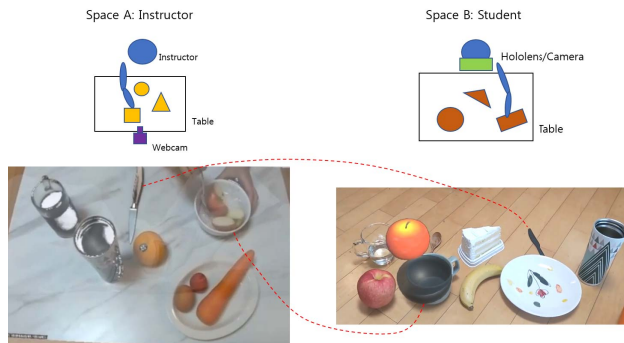


Figure 4: The system set-up of Table2Table tele-presence system (above). Below shows an example scene with objects in space A possibly mapping to those in space B.

3.1 Recognizing Object and Matching

The first task in T2T is to establish a match between the objects in the instructor's and student's spaces. For now, a simple method, just for the demonstrating the overall concept of T2T, based on object/type recognition is used. Namely, objects in the respective spaces as seen from the webcams are recognized using the YOLO v5 [8], a deep learning based object recognizer which can return the object labels (one of 150) and their screen space location (see Figure 5). A more robust and refined method of object matching remains as future work.

The object's possible roles are pre-identified based on their type (for a given scenario, in this case, the cooking instruction with kitchen objects). Similar objects in the context of the scenario can be grouped, and this information is used to suggest an initial match.

Figure 4 shows a possible match between the instructor's and student's table. Since there can be multiple objects of the same type (as illustrated in Figures 6 and 7), they are simply referred to by number. The initially suggested match is presented to the user for one to edit and modify, if necessary. The AR user interface for this purpose and an example of the match modification are illustrated in Figures 6 and 7.

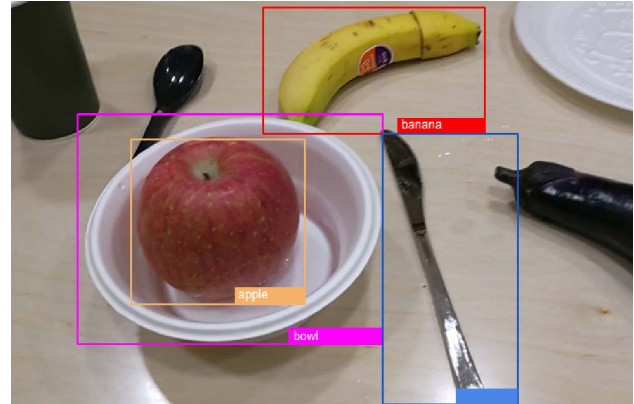


Figure 5: Kitchen objects recognized and their screen space locations using the YOLO v5 [8].

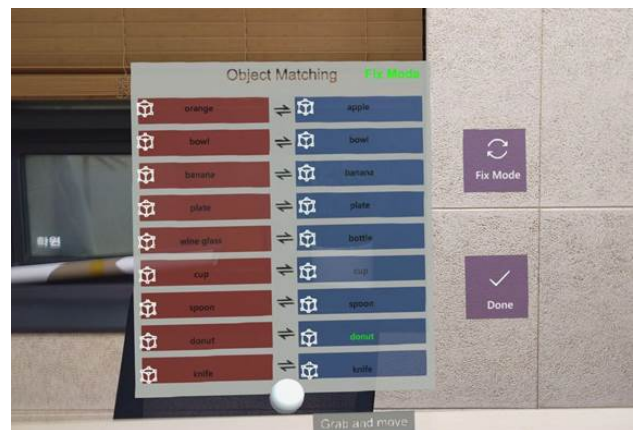


Figure 6: The AR user interface to modify the initially system suggested object match between the instructor's and student's table.

3.2 Recognizing Instructor's Actions

While the instructor gives the hands-on physical demonstration by manipulating various objects on one table, the action is recognized rather than captured (or recorded). Given that the action recognition is correct, its depiction in the form of 3D animation on the student's side can be made much clearer, e.g. than simply delivering the captured/recorded version which can sometimes be difficult to interpret. As the type of action is known, other useful information can be added as well. For instance, in illustrating the chopping of vegetables in a certain way, other similar or different use cases can be shown at the same time to facilitate the understanding.

The overall action recognition uses a simple state transition model. The system starts out from a default initial state and as events happen (mainly nominal object detection or movement tracking), the state updates according to simple rules. A state is basically a collection of facts represented in predicates parameterized by the instructor's hand and objects in the scene. The rules also specify under what conditions, new facts will be added and what specific actions to output and visualize. Objects involved in the instructor's action and one's hands are respectively recognized and grossly tracked using the same YOLO v5 (used in the object matching stage), and the Hololens MRTK [12]. Note that in this implementation of T2T, as the Hololens camera and MRTK are needed, the instructor has to wear the Hololens, even though its augmented display itself is not necessary nor used. In the future, the hand tracking module can be solely based on the webcam (as the YOLO object recognition).

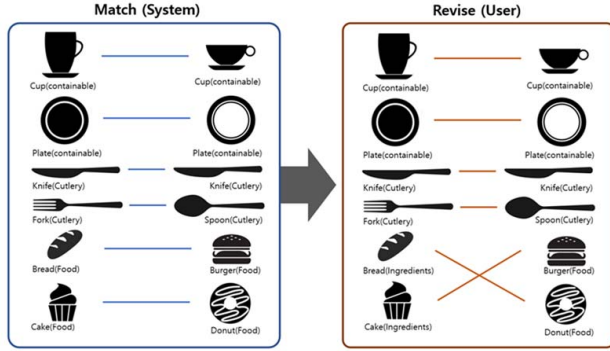


Figure 7: The initial match (left) is changed to the actual (right) by the user.

Objects and hands detected are bound to the parameters of the action recognition rules in searching for the satisfying conditions for rules to fire and carry out the state update. Again, for now just to demonstrate the overall concept of T2T, a simple rule-based approach is used to recognize the scenario-specific actions and assumptions (e.g. in this case, for cooking all driven by the hands) based on the hand location and movement pattern (e.g. grab, mix, pour), object proximity to the hand, and object types (tool, dish/bowl or food material). An illustrated example of how the action recognition works and a simple grammar for the action description are shown in Figure 8.

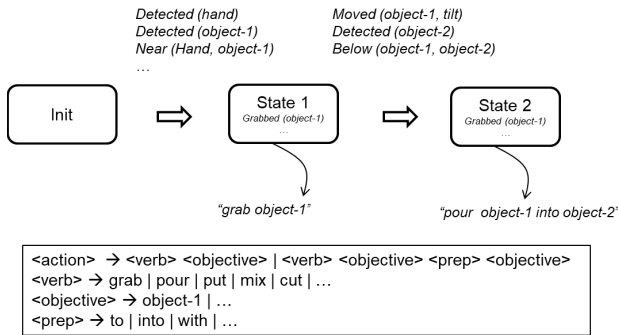


Figure 8: An illustrated example of maintaining “states” of the instructor and recognizing one’s action, in this case, “grab object-1” and “pour object-1 into object-2”. Below shows a simple grammar for the action description in the current implementation of T2T.

3.3 Augmenting Animation

Finally, the action description is used as an input to match and invoke pre-stored semi-parameterized 3D animations, and augment it to the objects involved in the action description. For instance, if the instructor’s action was “pour cup-1 into bowl-1” and the “cup-1” and “bowl-1” were matched respectively to “mug-1” and “pot-3” on the student’s space, an animation of a generic cup (pre-modeled) being tilted toward the “pot-3” will appear near the “mug-1”. Figure 9 shows examples of four such animated augmentations. Note that the augmentation is only for now roughly registered on the pertaining objects based on their screen space locations found during the object matching and action recognition phases.

The stored animations which are invoked are parameterized e.g. by the target objects being animated, source/destination locations and distance (for movement), type of movement (e.g. circular, linear, curved). As a proof of concept system, again only a handful of action animations are available like “grab”, “put”, “mix”, “cut”, “pour”, etc. The objects being animated are already pre-modeled in a generic

way for the given domain (in this case, cooking) like “cup”, “bowl”, “spoon”, “piece of meat”, “apple”, “bottle”, “knife”, etc.



Figure 9: Four examples of animated actions (from the top left in clockwise order): put object-1 onto object-2, pour object-1 into object-2, mix object-1 with object-2, and cut object-1 with object-2.

4 PILOT EXPERIMENT

We conducted a preliminary informal pilot experiment comparing the T2T system to the 2D video conferencing solution. Twenty subjects (mean age = 27.1, SD = 3.19) were asked to follow and perform a series of 10 kitchen tasks as guided by a live instructor using the respective systems: T2T and 2D Zoom [20]. Note that the tested T2T system did not include the optional video feed from the instructor’s web cam nor any voice instructions. Subjects (posing as students) wore the HoloLens for both conditions to remove any bias from wearing the display device. An example of a kitchen task was “put a carrot on a plate”, or like “cut a banana with a knife”. The examples above represent the level of complexity of the instructions (with just six possible verbs and 10 different object types).

As our assumption, the tables of the instructor and student were placed with similar objects in different locations. The task completion time, accuracy and general usability were measured. As for the task completion time, the time taken, once the instruction was given and animation shown, to mimic the task (correctly or not) was measured, and totaled for the whole sequences. Accuracy refers to whether the right objects were manipulated in the right way in following the instruction. The general usability was measure by a simplified questionnaire assessing ease of use, understandability, and instructor’s presence. We were able to observe that in both accuracy and task completion time, there were statistically significant differences: namely, when using the 2D video conferencing, the accuracy of the completed task was less (p-value=0.00249), and their time longer (p-value ≤ 0.0000).

We posit that this is due to the confusion and mental load on the part of the student in terms of locating the right object in mimicking the instructor. No particular difference was found on general usability, however, subjects reported of less presence felt, with T2T, with regard to the instructor. Note that the 3D animation does not include the appearance of the instructor, while it would be well presented through the 2D video conferencing. Thus as future work, we hope to extend the system to include at least some body parts of the instructor (as showing the whole figure could unnecessarily occlude the instructional scene) [13] [17] [19]. While most subjects confirmed the ease in understanding the instruction as it was replayed in their perspective over the matched objects, they also complained of the unnaturalness of the 3D animation.

5 CONCLUSION

In this paper, we proposed a new tele-presence framework between two similar table spaces, in which instead of a direct capture, the actions in one space is recognized and re-enacted in the other, by semi-automatically matching the objects between the spaces. The pilot experiment with the proof-of-concept implementation, T2T, demonstrated its potential advantage over the conventional 2D video conferencing, reducing the mental effort and confusion for having to match objects between the two spaces. By relying on recognition rather than capture, the overall system set-up becomes much simpler, and the 3D animated augmentation also allows for more flexible variation on the instruction (e.g. certain part of the instruction can easily be exaggerated, or unnecessary parts removed). There are many parts of the system that need further improvements including the schemes for object matching, action recognition and animation generation, particularly for applying the concept to a more practical and real situations. Another avenue for future work also include the use of voice in assisting the recognition of and re-enacted the instructor's actions.

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